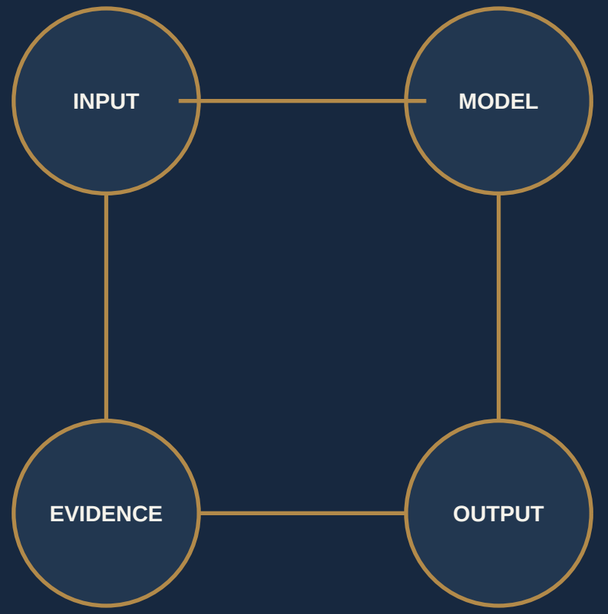
BACHELOR OF ARTIFICIAL INTELLIGENCE & ROBOTICS ENGINEERING

Deep Learning

AI.210 | University lecture series



What you will learn

A practical progression from core ideas to engineering judgment

- Explain and define cnns.
- Apply key methods including rnns.
- Build or analyze a practical system involving lstms.
- Evaluate assumptions, limitations, performance and responsible use.

A connected learning journey

Each stage prepares the tools needed for the next.

01

FOUNDATIONS

CNNs

02

METHODS

RNNs

03

SYSTEMS

LSTMs

04

PRACTICE

Laboratory, evaluation and project

The subject at a glance

A map of the ideas and techniques in this course

01 · CNNs
CNNs

02 · RNNs
RNNs

03 · LSTMs
LSTMs

04 · Attention
Attention

05 · Transformers
Transformers

06 · Optimization
Optimization

07 · Regularization
Regularization

08 · Transfer learning and model de
Transfer learning and model deployment

Core ideas, made explicit

AI.210 • Distinguish definitions, assumptions and evidence

- CNNs
- RNNs
- LSTMs
- Attention
- Transformers

DEFINE → MODEL → TEST → INTERPRET

Core ideas, made explicit

AI.210 • Distinguish definitions, assumptions and evidence

- Optimization
- Regularization
- Transfer learning and model deployment
- Mandatory practicals: Train deep-learning models using real datasets and deploy at least one practical application

DEFINE → MODEL → TEST → INTERPRET

Follow the system through its stages

Use the process diagram to connect components and outcomes.

PROBLEM

CNNs

INPUTS

RNNs

METHOD

LSTMs

OUTPUT

Attention

EVALUATION

Evidence, constraints and limitations

Teaching sequence: reveal one component at a time; then replay the complete end-to-end flow.

From concept to a real engineering decision

A short case study for discussion and worked-example narration

SCENARIO

A real task involving cnns.

CONSTRAINTS

Data, resources, safety and operating conditions shape the design.

DECISION

Select a method; justify the trade-offs and define success.

- State assumptions before calculating or implementing.
- Compare the expected result with observed evidence.

Laboratory · build, measure, improve

Students work through the following guided sequence.

- Specify a measurable task using cnns.
- Implement or simulate a baseline using rnns.
- Run a controlled test and record results with units and conditions.
- Diagnose a failure, revise the design and explain the improvement.

Review · explain · apply

KNOWLEDGE CHECK

What assumptions does the method make? What evidence would show that it works?

PRACTICAL CHALLENGE

Apply one course idea to a new context; state constraints, test criteria and likely failure modes.

Key takeaway: make the model visible, test it, and communicate its limits.

Course-series slide design • detailed narration, demonstrations and assessments expand each module.